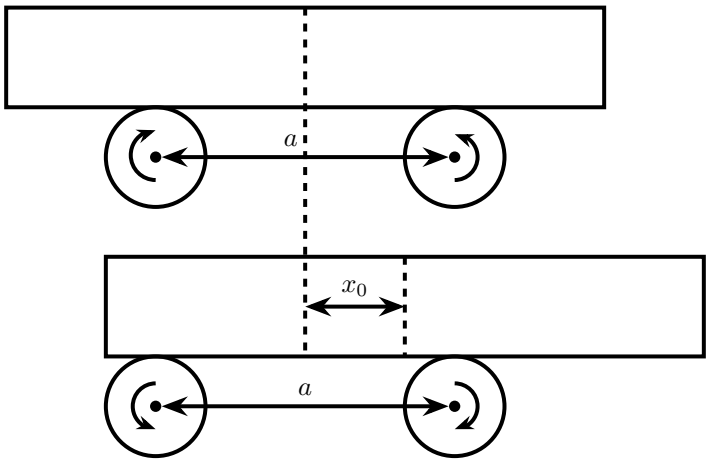
Physics Olympiad L2-22

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Problem 1: **Rolling Plank**

A uniform thin rigid rod of mass M is supported by two rotating rollers whose axes are separated by a fixed distance a. The rod is initially placed at rest asymmetrically at a distance x_0 away from the equilibrium position. The coefficient of friction between the bar and the rollers is μ. Determine the displacement of the centre of mass of the rod $x(t)$ when:

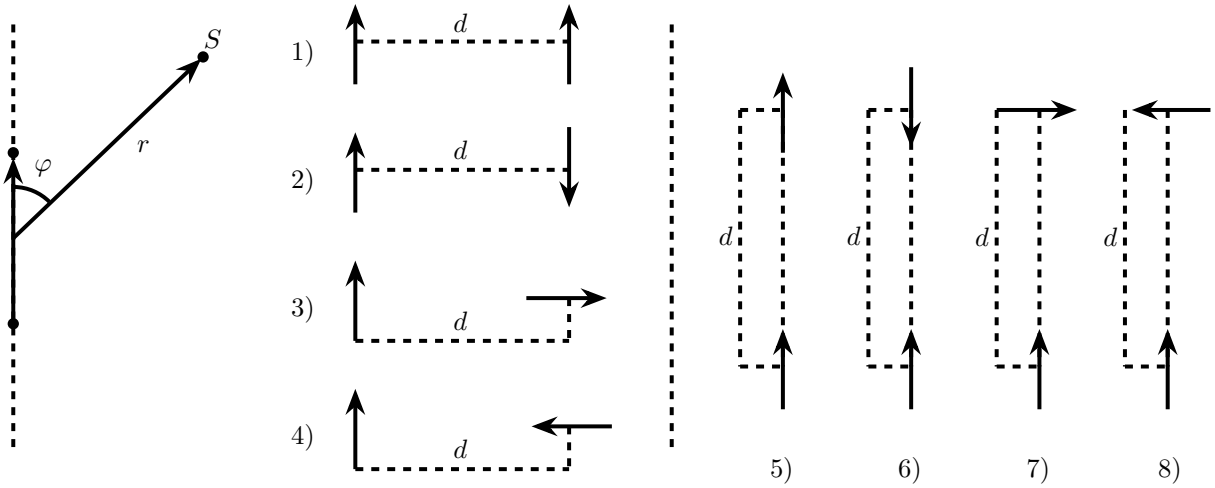
- The rollers rotate in opposite direction as shown in the figure;
- The rollers rotate in reverse directions to the ones shown in the figure;



Problem 2: **Interacting Dipoles**

Assume you are given a dipole with known charge q and length l, or equivalently the dipole moment $\vec{p}$.

- Find an expression for the potential φ of the dipole at point S as shown on the figure below;
- Using this expression, find the polar components of the electric field strength vector;
- Find the points at which $\vec{E}$ is perpendicular to $\vec{p}$;
- The dipole is placed in an external uniform electric field whose strength is $\overrightarrow{E_0} \uparrow\uparrow \vec{p}$. In this case one of the equipotential surfaces enclosing the dipole forms a sphere. Determine the radius of this sphere;
- In figures 1) through 8) two identical dipoles are placed in different arrangements as shown, with their centres being a distance $d >> l$ from one another in each case. For each arrangement shown find approximated expressions for the energy of interaction E_p and the force of interaction $\vec{F}$.



Problem 3: **Wave Superposition**

Two plane waves ψ_1 and ψ_2 propagate without energy loss in a homogeneous elastic medium, one along the x-axis and the other along the y-axis:

$\rightarrow\ \psi_1(x,t) = a\cos(\omega t - kx)$;

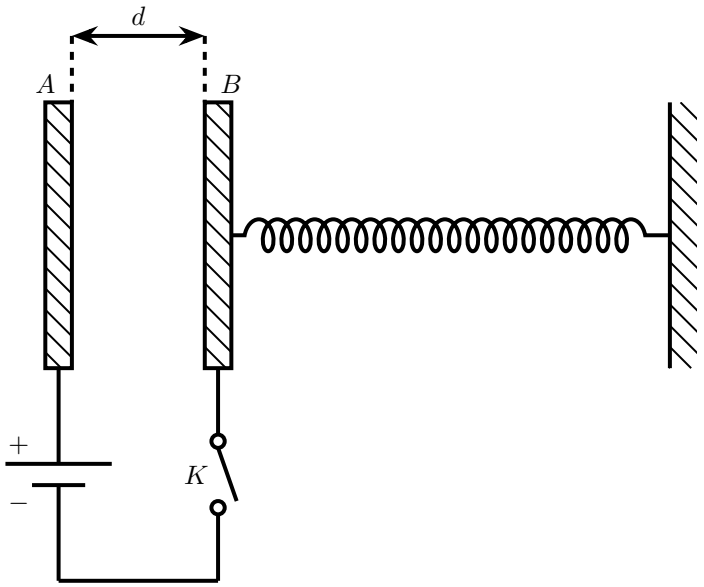
$\rightarrow\ \psi_2(y,t) = a\cos(\omega t - ky)$;

Find the wave motion pattern of the particles in the plane xy if both waves are:

- transverse and their oscillation directions coincide;
- longitudinal.

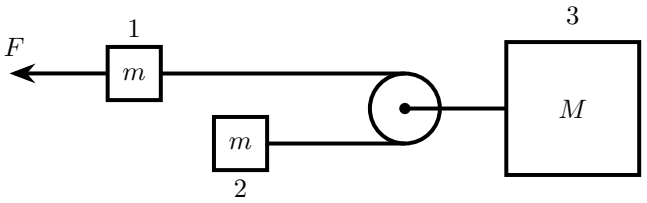
Problem 4: **Oscillating Capacitor**

Plate A of a parallel-plate capacitor is fixed, while plate B is attached to a wall with a spring and can move while remaining parallel to plate A. After closing the switch K, plate B starts moving and eventually stops in a new equilibrium position. In this case, the initial equilibrium distance d between the capacitor plates (when the spring is not stretched) decreases by $\eta = 10\%$. How much would the equilibrium distance between the plates change if the switch K were closed for a short time? Assume that during this time, plate B does not have time to move significantly.



Problem 5: **Body-Pulley System**

Find the acceleration of the bodies in the system shown in the figure. The force F is applied along the direction of the string to one of the bodies of mass m. The sections of the string on both sides of the light pulley, which is attached to a body of mass M, are parallel.



Problem 6: **Battery Efficiency**

A battery consists of N identical cells, each of e.m.f. ε. Is it true that the energy wasted when using the battery to charge a capacitor through a resistor can be reduced by charging it in N stages? That is by connecting it first across a single cell, and then across two cells, and so on, rather than across the whole battery in a single step.

Appendix

Useful Mathematical Formulae and Physical Constants

Important: Infinite series that are marked with $(*)$ are only valid under the condition that $|x| < 1$.

Taylor Expansions

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(x^7)$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \mathcal{O}(x^8)$$

$$\mathrm{tg}(x) = x + \frac{x^3}{3} + \frac{2x^5}{15} + \mathcal{O}(x^7)$$

$$\arcsin(x) = x + \frac{x^3}{6} + \frac{3x^5}{40} + \mathcal{O}(x^7)$$

$$\mathrm{arctg}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} + \mathcal{O}(x^7)$$

$$\exp(x) = 1 + x + \frac{x^2}{2} + \mathcal{O}(x^3)$$

$$\ln(1+x) = x - \frac{1}{2}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1-x)^{-1} = 1 + x + x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-1} = 1 - x + x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{+\frac{1}{2}} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-\frac{1}{2}} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{+\frac{3}{2}} = 1 + \frac{3}{2}x + \frac{3}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

$$(1+x)^{-\frac{3}{2}} = 1 - \frac{3}{2}x + \frac{15}{8}x^2 + \mathcal{O}(x^3) \qquad (*)$$

Inverse Hyperbolic Functions

$$\mathrm{arsinh}(x) = \ln(x + \sqrt{x^2+1})$$

$$\mathrm{arcosh}(x) = \ln(x + \sqrt{x^2-1})$$

$$\mathrm{artgh}(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

$$\mathrm{arctgh}(x) = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$$

Indefinite Integrals

$$\int (1-x^2)^{-\frac{3}{2}}\,dx = \frac{x}{\sqrt{1-x^2}} + C$$

$$\int (1+x^2)^{-\frac{3}{2}}\,dx = \frac{x}{\sqrt{1+x^2}} + C$$

$$\int (1-x^2)^{-\frac{1}{2}}\,dx = \arcsin(x) + C$$

$$\int (1+x^2)^{-\frac{1}{2}}\,dx = \mathrm{arsinh}(x) + C$$

$$\int (1-x^2)^{+\frac{1}{2}}\,dx = \frac{\arcsin(x) + x\sqrt{1-x^2}}{2} + C$$

$$\int (1+x^2)^{+\frac{1}{2}}\,dx = \frac{\mathrm{arsinh}(x) + x\sqrt{1+x^2}}{2} + C$$

Useful Physical Constants

Constant	Symbol	Value	Measuring Units
Light speed in vacuum	c	$3{,}000 \times 10^{8}$	m/s
Rest mass of electron	m_e	$9{,}109 \times 10^{-31}$	kg
Rest mass of proton	m_p	$1{,}673 \times 10^{-27}$	kg
Rest mass of neutron	m_n	$1{,}675 \times 10^{-27}$	kg
Rest mass of muon	m_μ	$1{,}884 \times 10^{-28}$	kg
Rest mass of tau-lepton	m_τ	$3{,}168 \times 10^{-27}$	kg
Elementary charge	e	$1{,}602 \times 10^{-19}$	C
Planck's constant	h	$6{,}626 \times 10^{-34}$	Js
Boltzmann's constant	k	$1{,}381 \times 10^{-23}$	J/K
Gravitational constant	γ	$6{,}673 \times 10^{-11}$	$\mathrm{Nm^2/kg^2}$
Dielectric permittivity of vacuum	ε_0	$8{,}854 \times 10^{-12}$	F/m
Avogadro's number	N_A	$6{,}022 \times 10^{23}$	$\mathrm{mol^{-1}}$
Rydberg's constant	R_∞	$1{,}097 \times 10^{7}$	$\mathrm{m^{-1}}$
Magnetic permeability of vacuum	μ_0	$4\pi \times 10^{-7}$	H/m
Universal gas constant	R	8,314	$\mathrm{J/(mol \cdot K)}$